DATA PRODUCT CASE STUDY · STELLARALGO

One source of truth for a multi-tenant product

How I made internal reporting consistent and reliable, at scale.

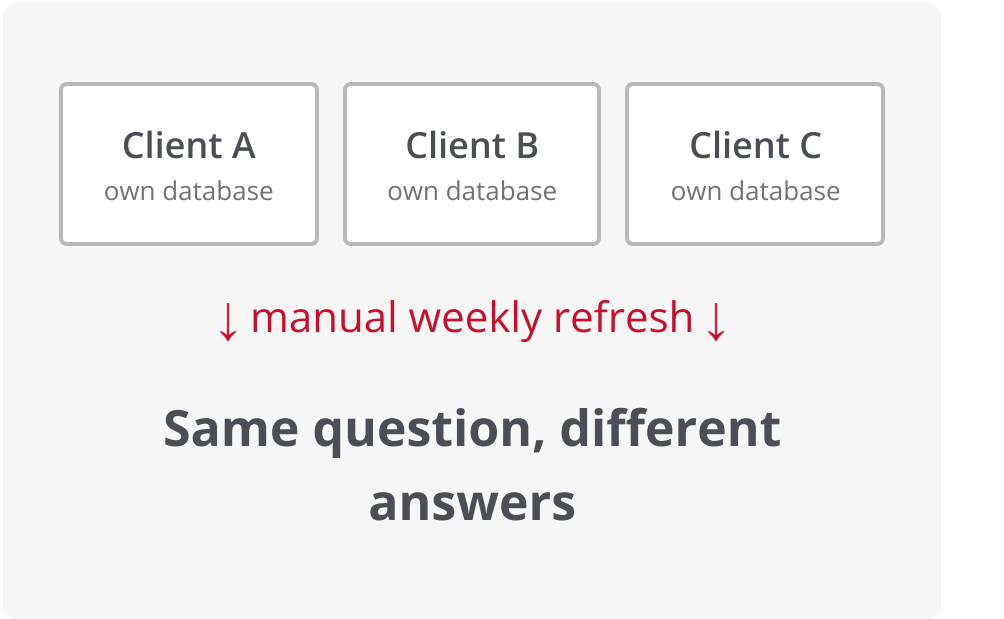
Cody Chandler
Data and Business Analyst interview · The City of Calgary

I'll answer them in order

1. What business problem was I solving?
2. What is the data product?
3. How did I approach it?
4. What challenges came up?
5. What am I most proud of?
6. What would I improve with more time?

What business problem was I solving?

The problem to solve: no single analytical source of truth, so reporting couldn't be consistent or reliable at scale.



Each client's data lived in its own database, so accurate internal reporting was hard. Reports were up to **7 days old** and nobody owned the numbers.

a **Why it mattered**

Teams worked from different numbers, and every cross-client question needed engineering.

b **Users and interested parties**

Customer Success · Product · Clients · Analysts

What is the data product?

a One governed, tested layer for every client's data, refreshed daily.

SOURCES
Three systems
Every client's data

→

BRONZE
Raw
Personal data masked

→

SILVER
One definition
Across every client

→

GOLD
Tested tables
One grain each

→

SEMANTIC LAYER
Shared metrics
For people and AI

b **Used by**

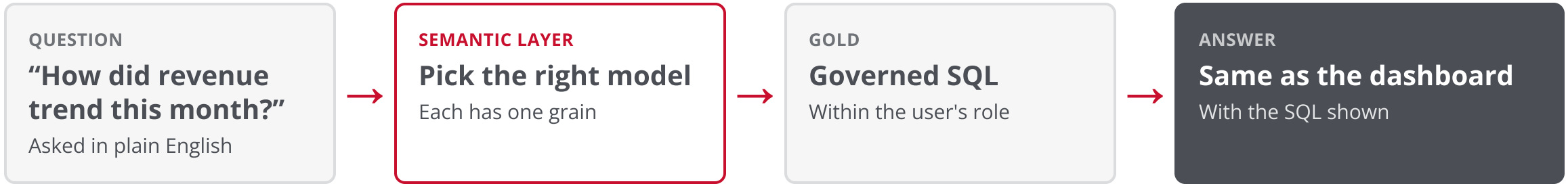
BI dashboards
Internal reporting

Client data sharing
Each client's own data

AI assistant
Plain-English questions

ML models
Trained on tested data

b How people use it: ask in plain English



Each model has one grain, so the AI can't double count. It reads the same truth as BI.

C What value it delivered

~~7 days~~

< 12 hrs

Data freshness

~~Per client~~

1 layer

Governed source of truth for every client

30 min

To flag a missed refresh, before a client notices

Self-serve

Routine Customer Success questions

Every team now reports from one source of truth, on a schedule they can rely on.

How did I approach it?

Requirements
What each team needed to trust a number.

→

Strategy
Fix the data once, not every report.

→

Plan
Bronze, silver, gold. One grain per table.

→

Build
Airflow and dbt, daily instead of weekly.

→

Review and iterate
Monitor, find issues, fix them once.

a **Sourcing and preparing the data**

Three source systems, mapped client by client into one model, then tested.

b **Methods and tools**

Snowflake · dbt · Airflow · AWS · Semantic layer · LLMs

What challenges came up, and how did I handle them?

Type	Challenge	How I solved it
Technical	Every client system had its own shape.	One model all clients map into.
Analytical	The same metric meant different things to different teams.	One definition, one documented grain per table.
People	Nobody owned freshness or accuracy.	A data contract I wrote: owners, targets and incident tiers.

What am I most proud of?

Governance stopped being overhead and became the reason people could self-serve.

Dashboards, client data and the AI all read from one source of truth, so they all give the same answer.

What would I improve with more time?

Measure from day one

Baseline hours and escalations before launch, so value is a number.

Take it to clients

Offer the internal AI assistant to clients, on the same layer.

Cover more of the business

More semantic models, each with one grain.

The same work at the City: find where numbers disagree, agree one definition with the owners, and make it self-serve.

The same approach, live on City of Calgary data

1. Approach: the pipeline and its lineage
2. Dashboard: a 43% jump that wasn't real
3. Predict: a live machine learning model
4. Ask the data: plain English, with its working

codychandler.ca/data-demos/calgary-transit

Thank you. Questions?

Live demo

codychandler.ca/data-demos/calgary-transit